

# LEXAN<sup>TM</sup> COPOLYMER EXL9414

REGION ASIA

## DESCRIPTION

LEXAN EXL9414 polycarbonate (PC) siloxane copolymer resin is a medium flow, non-chlorinated, non-brominated flame retardant opaque injection molding (IM) grade. This resin offers low temperature ductility, thin wall flame retardant capability, and in combination with excellent processability and release with opportunities for shorter IM cycle times compared to standard PC. LEXAN EXL9414 resin is a product available in wide range of opaque colors and may be an excellent candidate for a wide variety of applications, especially the housing of fast-charging mobile phones.

## TYPICAL PROPERTY VALUES

Revision 20200423

| PROPERTIES                                   | TYPICAL VALUES | UNITS             | TEST METHODS |
|----------------------------------------------|----------------|-------------------|--------------|
| <b>MECHANICAL</b>                            |                |                   |              |
| Tensile Stress, yld, Type I, 50 mm/min       | 56             | MPa               | ASTM D 638   |
| Tensile Stress, brk, Type I, 50 mm/min       | 62             | MPa               | ASTM D 638   |
| Tensile Strain, yld, Type I, 50 mm/min       | 5.8            | %                 | ASTM D 638   |
| Tensile Strain, brk, Type I, 50 mm/min       | 107            | %                 | ASTM D 638   |
| Tensile Modulus, 50 mm/min                   | 2110           | MPa               | ASTM D 638   |
| Flexural Stress, yld, 1.3 mm/min, 50 mm span | 88             | MPa               | ASTM D 790   |
| Flexural Modulus, 1.3 mm/min, 50 mm span     | 2190           | MPa               | ASTM D 790   |
| Hardness, Rockwell L                         | 83             | -                 | ASTM D 785   |
| Hardness, Rockwell R                         | 117            | -                 | ASTM D 785   |
| Tensile Stress, yield, 50 mm/min             | 57             | MPa               | ISO 527      |
| Tensile Stress, break, 50 mm/min             | 60             | MPa               | ISO 527      |
| Tensile Strain, yield, 50 mm/min             | 5.6            | %                 | ISO 527      |
| Tensile Strain, break, 50 mm/min             | 106            | %                 | ISO 527      |
| Tensile Modulus, 1 mm/min                    | 2140           | MPa               | ISO 527      |
| Flexural Stress, yield, 2 mm/min             | 86             | MPa               | ISO 178      |
| Flexural Modulus, 2 mm/min                   | 2180           | MPa               | ISO 178      |
| <b>IMPACT</b>                                |                |                   |              |
| Izod Impact, notched, 23°C                   | 880            | J/m               | ASTM D 256   |
| Izod Impact, notched, -30°C                  | 660            | J/m               | ASTM D 256   |
| Instrumented Impact Total Energy, 23°C       | 67             | J                 | ASTM D 3763  |
| Izod Impact, notched 80*10*3 +23°C           | 69             | kJ/m <sup>2</sup> | ISO 180/1A   |
| Izod Impact, notched 80*10*3 -30°C           | 46             | kJ/m <sup>2</sup> | ISO 180/1A   |
| Charpy 23°C, V-notch Edgew 80*10*3 sp=62mm   | 75             | kJ/m <sup>2</sup> | ISO 179/1eA  |
| Charpy -30°C, V-notch Edgew 80*10*3 sp=62mm  | 29             | kJ/m <sup>2</sup> | ISO 179/1eA  |
| <b>THERMAL</b>                               |                |                   |              |
| HDT, 0.45 MPa, 3.2 mm, unannealed            | 134            | °C                | ASTM D 648   |
| HDT, 1.82 MPa, 3.2mm, unannealed             | 118            | °C                | ASTM D 648   |
| CTE, -40°C to 40°C, flow                     | 6.91E-05       | 1/°C              | ASTM E 831   |
| CTE, -40°C to 40°C, xflow                    | 7.27E-05       | 1/°C              | ASTM E 831   |
| CTE, 23°C to 80°C, flow                      | 7.42E-05       | 1/°C              | ISO 11359-2  |
| CTE, 23°C to 80°C, xflow                     | 7.64E-05       | 1/°C              | ISO 11359-2  |
| Vicat Softening Temp, Rate B/50              | 136            | °C                | ISO 306      |

| PROPERTIES                                                   | TYPICAL VALUES                    | UNITS      | TEST METHODS   |
|--------------------------------------------------------------|-----------------------------------|------------|----------------|
| Vicat Softening Temp, Rate B/120                             | 138                               | °C         | ISO 306        |
| HDT/Be, 0.45MPa Edgew 120*10*4 sp=100mm                      | 131                               | °C         | ISO 75/Be      |
| HDT/Ae, 1.8 MPa Edgew 120*10*4 sp=100mm                      | 117                               | °C         | ISO 75/Ae      |
| Relative Temp Index, Elec <sup>(1)</sup>                     | 80                                | °C         | UL 746B        |
| Relative Temp Index, Mech w/impact <sup>(1)</sup>            | 80                                | °C         | UL 746B        |
| Relative Temp Index, Mech w/o impact <sup>(1)</sup>          | 80                                | °C         | UL 746B        |
| PHYSICAL                                                     |                                   |            |                |
| Specific Gravity                                             | 1.19                              | -          | ASTM D 792     |
| Mold Shrinkage, flow, 3.2 mm                                 | 0.4 – 0.8                         | %          | SABIC method   |
| Mold Shrinkage, xflow, 3.2 mm                                | 0.4 – 0.8                         | %          | SABIC method   |
| Melt Flow Rate, 300°C/1.2 kgf                                | 11.5                              | g/10 min   | ASTM D 1238    |
| Density                                                      | 1.19                              | g/cm³      | ISO 1183       |
| Melt Volume Rate, MVR at 300°C/1.2 kg                        | 10                                | cm³/10 min | ISO 1133       |
| ELECTRICAL                                                   |                                   |            |                |
| Volume Resistivity                                           | >1.E+16                           | Ohm-cm     | ASTM D 257     |
| Surface Resistivity                                          | >1.E+16                           | Ohm        | ASTM D 257     |
| Dielectric Constant (Dk), 1.1 GHz                            | 2.78                              | -          | ASTM ES 7-83   |
| Dissipation Factor (Df), 1.1 GHz                             | 0.006                             | -          | ASTM ES 7-83   |
| Dielectric Strength, in oil, 3.2 mm                          | 13                                | kV/mm      | ASTM D 149     |
| FLAME CHARACTERISTICS <sup>(1)</sup>                         |                                   |            |                |
| UL Yellow Card Link                                          | <a href="#">E207780-102896543</a> | -          | -              |
| UL Recognized, 94-5VA Flame Class Rating                     | ≥3.1                              | mm         | UL 94          |
| UL Recognized, 94-5VB Flame Class Rating                     | ≥1.5                              | mm         | UL 94          |
| UL Recognized, 94V-0 Flame Class Rating                      | ≥1                                | mm         | UL 94          |
| UL Recognized, 94V-1 Flame Class Rating                      | ≥0.8                              | mm         | UL 94          |
| UL Recognized, 94V-2 Flame Class Rating                      | ≥0.5                              | mm         | UL 94          |
| Glow Wire Flammability Index 960°C, passes at <sup>(2)</sup> | 1                                 | mm         | IEC 60695-2-12 |
| Oxygen Index (LOI)                                           | 39                                | %          | ISO 4589       |
| INJECTION MOLDING                                            |                                   |            |                |
| Drying Temperature                                           | 120                               | °C         |                |
| Drying Time                                                  | 3 – 4                             | hrs        |                |
| Drying Time (Cumulative)                                     | 48                                | hrs        |                |
| Maximum Moisture Content                                     | 0.02                              | %          |                |
| Melt Temperature                                             | 295 – 315                         | °C         |                |
| Nozzle Temperature                                           | 290 – 310                         | °C         |                |
| Front - Zone 3 Temperature                                   | 295 – 315                         | °C         |                |
| Middle - Zone 2 Temperature                                  | 280 – 305                         | °C         |                |
| Rear - Zone 1 Temperature                                    | 270 – 295                         | °C         |                |
| Mold Temperature                                             | 70 – 95                           | °C         |                |
| Back Pressure                                                | 0.3 – 0.7                         | MPa        |                |
| Screw Speed                                                  | 40 – 70                           | rpm        |                |
| Shot to Cylinder Size                                        | 40 – 60                           | %          |                |
| Vent Depth                                                   | 0.025 – 0.076                     | mm         |                |



(1) UL Ratings shown on the technical datasheet might not cover the full range of thicknesses and colors. For details, please see the UL Yellow Card.

(2) Value shown here is based on internal measurement.

## DISCLAIMER

Any sale by SABIC, its subsidiaries and affiliates (each a "seller"), is made exclusively under seller's standard conditions of sale (available upon request) unless agreed otherwise in writing and signed on behalf of the seller. While the information contained herein is given in good faith, SELLER MAKES NO WARRANTY, EXPRESS OR IMPLIED, INCLUDING MERCHANTABILITY AND NONINFRINGEMENT OF INTELLECTUAL PROPERTY, NOR ASSUMES ANY LIABILITY, DIRECT OR INDIRECT, WITH RESPECT TO THE PERFORMANCE, SUITABILITY OR FITNESS FOR INTENDED USE OR PURPOSE OF THESE PRODUCTS IN ANY APPLICATION. Each customer must determine the suitability of seller materials for the customer's particular use through appropriate testing and analysis. No statement by seller concerning a possible use of any product, service or design is intended, or should be construed, to grant any license under any patent or other intellectual property right.